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Doppler-Free Laser Spectroscopy on Rubidium

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Abstract

Doppler-free saturated absorption spectroscopy was performed on rubidium vapour to investigate the hyperfine structure of the D_2 transition at 780 nm. Counter-propagating pump and probe beams from a DFB diode laser enabled resolution of individual hyperfine transitions in the ^{85}Rb and ^{87}Rb isotopes. A Fabry–Pérot cavity calibrated the frequency sweep, converting oscilloscope time into frequency. Gaussian fits to the Doppler-broadened profiles yielded magnetic-dipole coupling constants in the $5^2S_{1/2}$ ground-state of $A_s^{85} = 1005.7 \pm 15.1$MHz and $A_s^{87} = 3416.1 \pm 5.1$MHz. Lorentzian fits to the Doppler-free resonances gave $5^2P_{3/2}$ excited-state hyperfine constants consistent with literature values: $A_p^{85} = 25.9 \pm 1.3$MHz, $A_s^{87} = 86.7 \pm 2.7$MHz, $B_p^{85} = 28.4 \pm 2.8$MHz, and $B_p^{87} = 14.6 \pm 2.9$MHz.

Introduction

In this experiment, Doppler-free saturated absorption spectroscopy was applied to rubidium vapour on the D_2 transition near 780 nm. Precise spectroscopy of rubidium provides insight into the interactions between the valence electron and the nuclear magnetic dipole and electric quadrupole moments. These interactions result in fine structure in the $5^2S_{1/2}$ ground state and hyperfine structure in the $5^2P_{3/2}$ excited state.

Conventional absorption spectroscopy cannot resolve these splitting's in a thermal vapour because Doppler broadening arising from thermal motion (~ 510 MHz at room temperature) is almost two orders of magnitude larger than the transitions linewidths (~ 6 MHz). Doppler-free saturated absorption spectroscopy removes this inhomogeneous broadening by using counter-propagating pump and probe beams to isolate the population of zero-velocity atoms.

A distributed-feedback (DFB) diode laser was swept over a ~10GHz range covering all D_2 hyperfine regions. A Fabry–Pérot reference cavity provided frequency calibration by assuming a linear sweep between adjacent transmission resonances. Analysis of both the Doppler-broadened and Doppler-free spectra enabled measurement of the hyperfine splittings of ^{85}Rb and ^{87}Rb, from which the magnetic dipole and electric quadrupole coupling constants were extracted for the $5^2S_{1/2}$ and $5^2P_{3/2}$ states. These experimentally determined constants are compared with established literature values to demonstrate the high accuracy achievable with saturated absorption spectroscopy.

Theory

Hyperfine Structure

Rubidium has one valence electron, which is excited by the D_2 transition ($5^2\mathrm{S}_{1/2} \rightarrow 5^2\mathrm{P}_{3/2}$).
In its ground state ($J = {}^1/_2$), the total angular momentum is $F = I \pm ½$,
where I is the nuclear spin ($I = {}^5/_2$ for ^{85}Rb, and $I = {}^3/_2$ for ^{87}Rb).
When excited ($J = {}^3/_2$), the total angular momentum obeys the selection rule [1]

$$\Delta \mathrm{F} = 0, \pm 1 \quad (1).$$

These variations in total angular momentum produce hyperfine energy levels given by[2]

$$E = \frac{AK}{2} + b \cdot B \qquad (2).$$

(equation 2 shows hyperfine energy levels is a sum of magnetic dipole shift and the electric quadrupole shift)

Here, A is the magnetic dipole constant and B is the quadrupole constant.
The quantities K and b depend on the nuclear and electronic angular momenta:

$$K = F(F+1) - I(I+1) - J(J+1), \qquad b = \frac{(3/4)K(K+1) - I(I+1)J(J+1)}{2I(2I-1)J(2J-1)} \qquad (3a,b).$$

Doppler Broadening

Thermal motion in gaseous rubidium shifts the laser frequency via the Doppler effect. Atoms with a positive velocity component along the probe beam (v_z) observe a higher frequency (f') than the one emitted, $f' = f\left(1+\frac{v_z}{c}\right)$. An atom that absorbs at frequency f_0 in its rest frame therefore absorbs light in the laboratory frame when,

$$f - f_0 = f_0 \frac{v_z}{c} \qquad (4).$$

The velocity distribution along the beam due to thermal motion is Maxwell–Boltzmann:

$$f(v_z) \propto \exp\left(-\frac{mv_z^2}{2k_BT}\right), \qquad (5),$$

Because the Doppler shift is linear in v_z, this distribution leads to a Gaussian absorption line shape. The Doppler-broadened full width half maximum is, [1]

$$\Delta f_D = 2f_0\sqrt{\frac{2k_B \mathrm{T}\ln 2}{mc^2}} \qquad (6).$$

For the D_2 transition at 300K $\Delta f_D^{87} \approx$ 511MHz, and $\Delta f_D^{85} \approx$ 517MHz.
This is an example of inhomogeneous broadening, where atoms absorb at different apparent frequencies due to their velocity distribution.

These widths are enormous compared with the ~6 MHz natural linewidth of an individual hyperfine transition. As a result, absorption from atoms moving at different velocities spans a wide frequency range, causing all hyperfine transitions to merge into a single broad feature.

Doppler-Free Saturated Absorption

In saturated absorption spectroscopy a strong pump laser and a weak probe laser propagate through the rubidium vapour. These beams travel in opposite directions, hence an atom moving with v_z along the probe's axis sees two oppositely Doppler-shifted beams.
From equation 4, the pump and probe frequencies in the atom's rest frame become

$$f'_{probe} = f_0\left(1+\frac{v_z}{c}\right), \qquad f'_{pump} = f_0\left(1 - \frac{v_z}{c}\right) \qquad (7\mathrm{a,b}).$$

Only when the laser frequency is tuned exactly to the atomic resonance f_0 do atoms with $v_z \approx 0$ satisfy the resonance condition for both beams simultaneously. The strong pump beam is more probable to excite the population of $v_z \approx 0$ atoms, depleting the population of stationary atoms. This "hole-burning" increases the transmission of the probe beam at this frequency, producing a sharp Doppler-free *Lamb dip*. [1]

Lorentzian profiles

The Lamb dip produced by saturated absorption has a homogeneous Lorentzian line shape, whose width is determined by the intrinsic and power-broadened linewidth of the transition. The transmission profile of the probe beam near resonance is

$$I(\nu) \propto \frac{1}{(f' - f_0)^2 + \left(\Gamma_{eff}/2\right)^2} \quad (8).$$

Here $I(\nu)$ is the transmitted probe intensity, and Γ_{eff} is the power-broadened linewidth,

$$\Gamma_{eff} = \frac{1}{2\pi\tau}\sqrt{1 + \frac{I_{pump}}{I_{sat}}} \quad (9).$$

The $(2\pi\tau)^{-1}$ term gives the natural linewidth set by the excited-state lifetime τ, while the square-root factor accounts for power broadening, with I_{sat} defined as the intensity at which the excitation rate reaches half of its maximum (saturation intensity).

Crossover Resonances

Crossover resonances occur between any two hyperfine transitions within the same Doppler-broadened absorption profile. If two transitions lie at frequencies f_i and f_j, then a crossover peak appears at the midpoint frequency

$$f_{(i,j)} = \frac{f_i + f_j}{2} \quad (10).$$

At this frequency, atoms with a specific velocity simultaneously satisfy both resonance conditions (from Eq. 7a,b). Atoms with $+v_z$ Doppler-shift the pump into resonance with the higher-frequency transition f_j, while the probe is shifted into resonance with the lower-frequency transition f_i. Atoms with $-v_z$ satisfy the opposite pair. The pump therefore depletes the populations of atoms with velocities $+v_z$ and $-v_z$ at the same frequency, producing a Doppler-free hole-burning signal at the midpoint.

The contributions from atoms with velocities $\pm v_z$ add together, so the resulting crossover peak is nearly twice as strong as the individual hyperfine Lamb dips with only minor suppression for larger speeds due to the Maxwell–Boltzmann distribution.

The crossover amplitude approximately doubles, but the linewidth only increases by a factor of $\sqrt{2}$ due to statistical broadening. The variances add ($\sigma_{\mathrm{co}}^2 = 2\,\sigma_{\mathrm{hf}}^2$), giving $\sigma_{\mathrm{co}} = \sqrt{2}\,\sigma_{\mathrm{hf}}$, and hence the crossover linewidth satisfies

$$\Gamma_{\mathrm{co}} \approx \Gamma_{\mathrm{hf}}\sqrt{2} \quad (11).$$

Frequency Calibration

A Fabry–Pérot cavity was used to convert oscilloscope time into an absolute frequency axis. According to Yariv's cavity theory[3], the cavity transmits light when the round-trip phase shift satisfies the resonance conditions:

$$\psi_{phase} = 2\mathrm{m}\pi, \quad 2Ln = m\lambda \quad (12\mathrm{a,b}).$$

Where m is an integer number of half-wavelengths that fit inside the cavity.
The separation between successive resonances ($m \rightarrow m+1$) gives the free spectral range

$$\Delta f_{\mathrm{FSR}} = \frac{c}{2nL} \qquad (13).$$

These equally spaced resonances served as relative frequency markers. The oscilloscope recorded the peaks on a time axis, and their known separation in frequency allowed a linear interpolation between neighbouring cavity transmission peaks to convert time into frequency. This procedure yielded a calibrated frequency axis with an estimated uncertainty of $\sim 1.5\%$, arising from laser sweep non-linearity between cavity peaks

Statistical Weighting of Fitted Quantities

Throughout the analysis, fit parameters are extracted from Gaussian or Lorentzian models using the method of least squares. When multiple measurements of the same physical quantity are obtained, they are averaged using inverse-variance weighting so that measurements with smaller uncertainties have a greater contribution. For example, this is used to calculate the linewidth (Γ_{eff}) for the six measurements of ^{87}Rb hyperfine peaks.

The weighted mean of measured values x_i with uncertainties σ_i is defined using the weights

$$w_i = \frac{1}{\sigma_i^2}, \qquad \bar{x} = \frac{\Sigma_i w_i x_i}{\Sigma_i w_i}, \qquad \sigma_{\bar{x}} = \left(\Sigma_i w_i\right)^{-\frac{1}{2}} \qquad (14a, b, c).$$

This is applied at three stages of the analysis. It is first used to obtain the linewidths for both isotopes for both the hyperfine and crossover peaks. It is then used when determining the excited-state hyperfine constants A_p and B_p as we have two readings for each from the F_{low} and F_{high} ground levels. Finally, it is used to quantify the laser-sweep nonlinearity by combining the crossover-frequency deviations from all twelve crossover–hyperfine combinations.

Experimental Procedure

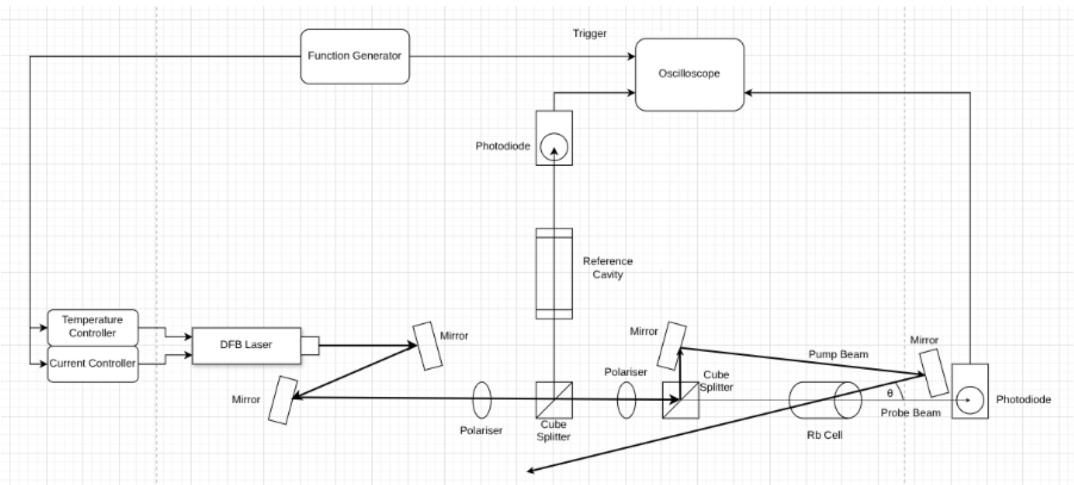

Figure 1 *presents the setup required for Doppler-Free Saturated Absorption Spectroscopy.*

A single-mode distributed-feedback (DFB) diode laser near 780 nm, passed through an isolator, polarisers and beam splitters to generate pump, probe, and reference beams. Photodiodes detected transmission signals simultaneously from a Fabry–Pérot cavity and the probe beam. A function generator drove laser current sweep, covering the minimum range (≈10GHz) for all D_2 transitions.

The laser beam was split into counter-propagating pump and probe paths that overlapped inside the vapour cell. To ensure the beams were as close to perfectly antiparallel as possible, the pump-reflecting mirror was positioned at the maximum practical distance from the cell. With a propagation distance of ≈50 cm and a lateral displacement of ≈5 mm, the resulting angular separation was reduced to $\theta \approx 0.6°$, to minimise Doppler-broadening.

Two polariser configurations were used during data collection. At $\vartheta = 0°$, the probe beam dominates, useful to record the Doppler-broadened absorption troughs. At $\vartheta = 73°$, approximately 91% of the intensity is directed into the pump beam, providing the configuration required for saturated absorption and producing clear hyperfine peaks.

Data Calibration

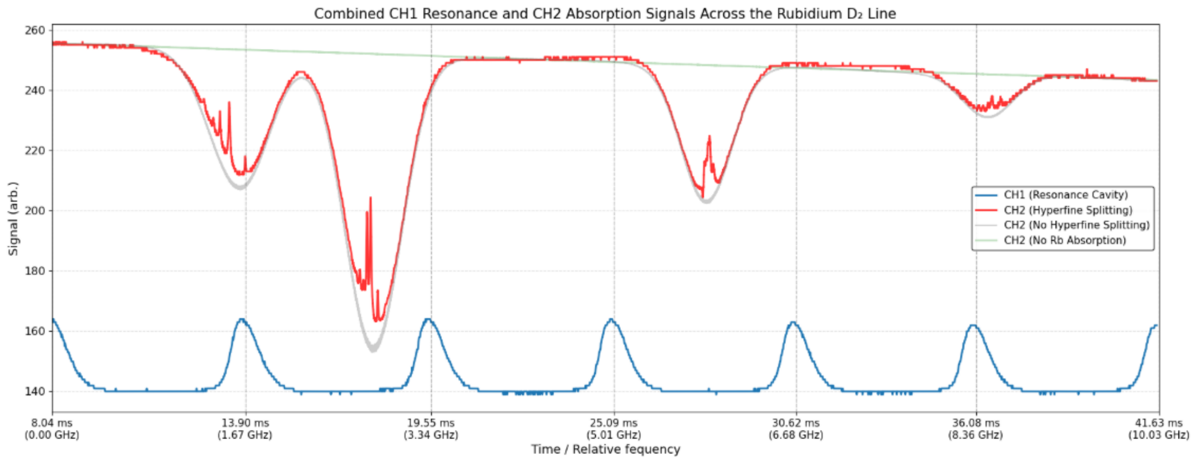

Figure 2 *presents the combined cavity calibration trace (blue), the Doppler-free saturated absorption spectrum (red), the Doppler-broadened absorption profile (grey), and the corresponding laser sweep function (green).*

A Gaussian profile for the Doppler-broadened absorption can be obtained by subtracting the Doppler-broadened signal from the laser sweep function. Similarly, the Doppler-free

Lorentzian hyperfine peaks can be baseline-corrected by subtracting the hyperfine spectrum from the Doppler-broadened profile.

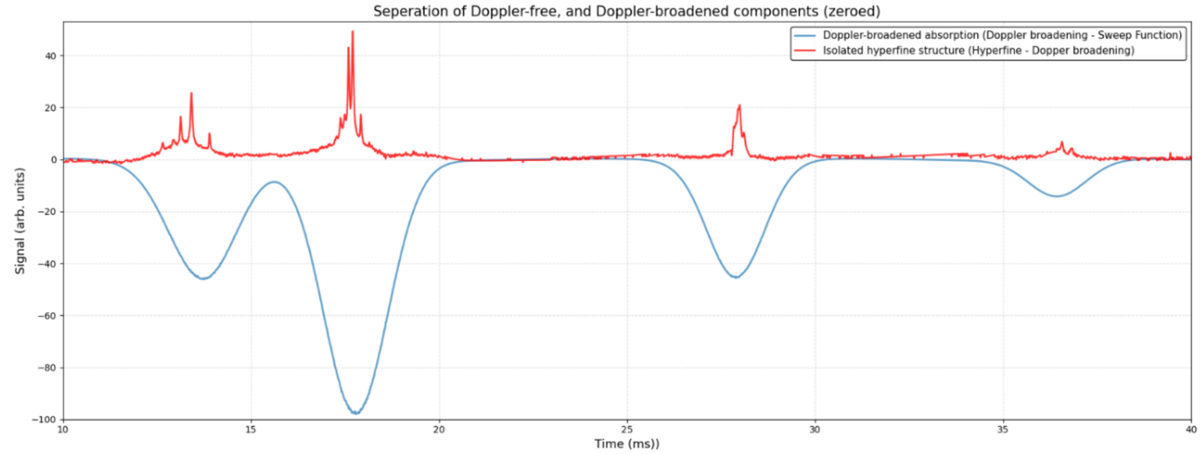

Figure 3 *shows the separation of the Doppler-broadened (blue) and Doppler-free (red) components.*

Using Eq. 13 and a reference cavity with refractive index n = 1.00 and length L = 0.0897 m, we determine the free spectral range to be Δf_{FSR} = 1.672 GHz. Frequency is assumed to scale linearly with oscilloscope time between cavity resonances; however, later results indicate that the effective sweep rate can deviate by ~1.5% due to non-linearities in the laser's response to the function generator.

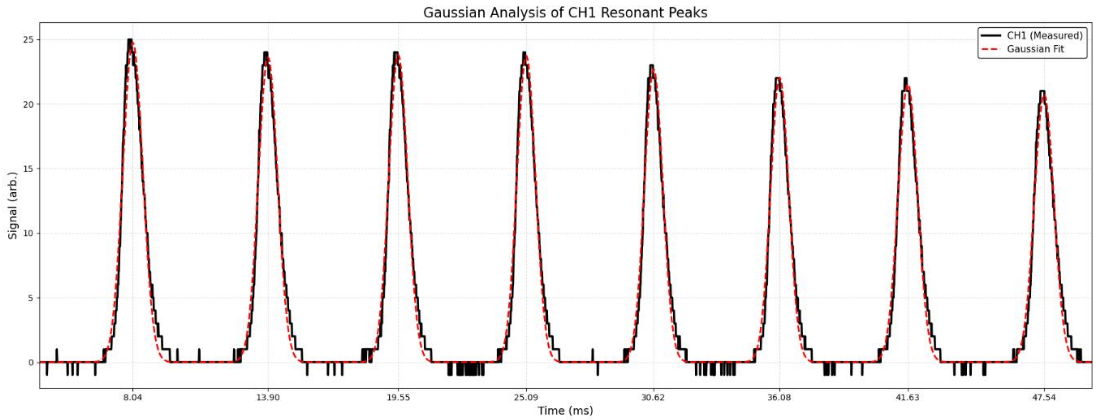

Figure 4 *shows the Gaussian fits applied to cavity resonant peaks ($R^2 = 0.984$), used to determine their centres*

Data Modelling

Doppler-Broadened Gaussian Model

After converting the data into the frequency domain, a Gaussian model is applied to the Doppler-broadened profile. This allows extraction of trough centres as well as parameters such as full width at half maximum (FWHM) and amplitude.

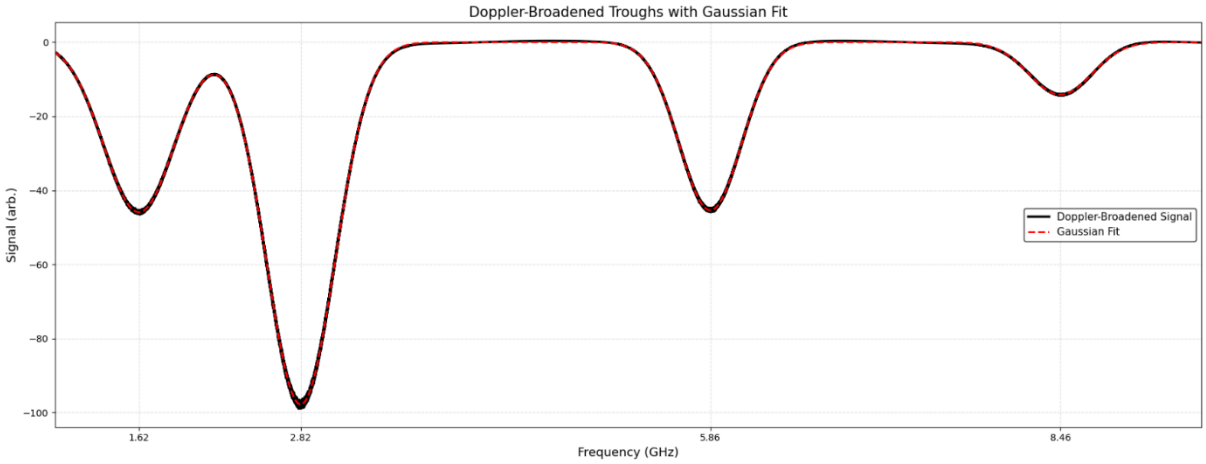

Figure 5 *shows the Doppler-broadened troughs in the frequency domain and with a Gaussian fit ($R^2 = 0.99991$).*

Doppler-Free Lorentzian Model

The Doppler-free spectrum separates into four distinct hyperfine regions, arising from the two rubidium isotopes, each with two different ground-state F-levels. Within each region, three hyperfine peaks are observed: the first peak corresponds to the $\Delta F = -1$ transition and is extremely weak, peak 3 corresponds to $\Delta F = 0$, and peak 6 to $\Delta F = +1$.

The three larger peaks are crossover resonances. Using the midpoint relation from Eq. 10, the crossovers appear as peak 2 at $f_{(1,3)}$ peak 4 at $f_{(1,6)}$, and peak 5 at $f_{(3,6)}$. These crossover resonances are noticeably broader and higher in amplitude due to the shared depletion of atoms with both velocities $\pm v_z$, making them the dominant features in the Doppler-free spectrum.

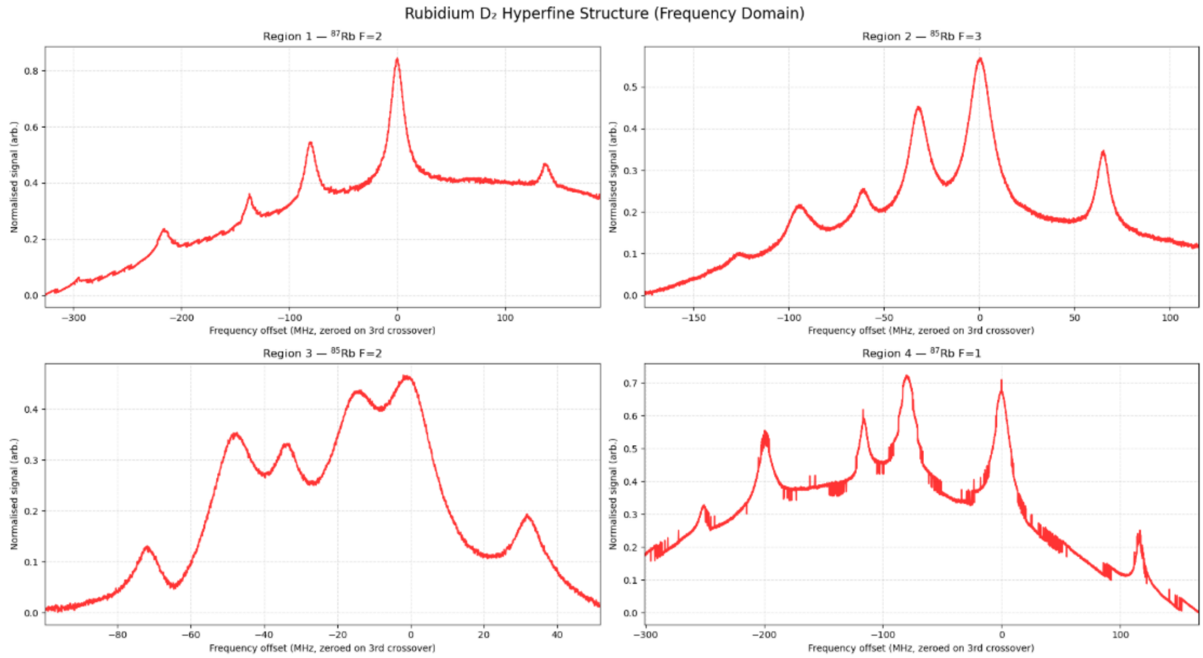

Figure 6 *displays the hyperfine regions for each isotope's valence electron at different ground-state F-levels.*

Using Python's *lmfit* module, the Doppler-free features are modelled as Lorentzian peaks superimposed on a polynomial background.

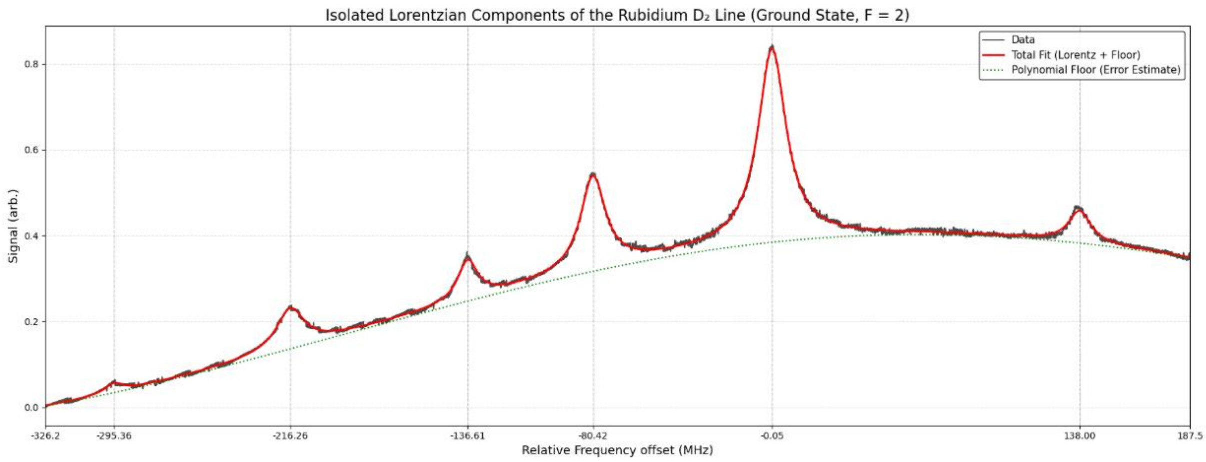

Figure 7 *shows the combined Lorentzian-plus-polynomial model (red) alongside the polynomial background (green).*

Subtracting this polynomial floor isolates the baseline-corrected Lorentzian hyperfine peaks.

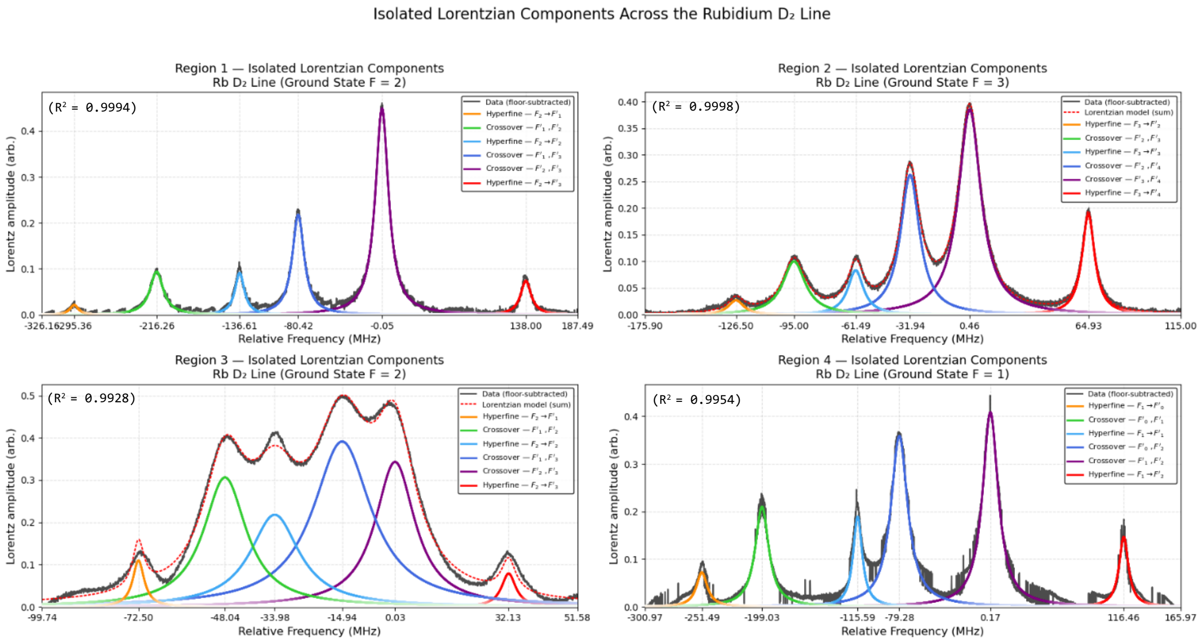

Figure 8 *shows the Lorentzian components across all hyperfine regions, with labelled hyperfine and crossover peaks.*

Analysis of Doppler-Broadened Troughs

Ground state energy splitting

The ground-state energy separations were extracted from the Gaussian fits to the Doppler-broadened absorption troughs.

For ⁸⁷Rb, the measured splitting between the $F = 1$ and $F = 2$ levels was found to be

$$\Delta E_{87} = 6.832 \text{ GHz}.$$

(within $\sim 1.5\%$ *sweep rate uncertainty of the published value* 6.835 *GHz)* [4]

For ⁸⁵Rb, the measured splitting between the $F = 2$ and $F = 3$ levels was found to be

$$\Delta E_{85} = 3.017 \text{ GHz},$$

(within $\sim 1.5\%$ *sweep rate uncertainty of the published value* 3.036 *GHz)* [5]

Doppler-Broadened FWHM

The full width half maximum of the Doppler models was obtained from Gaussian fits.

$$\text{FWHM}_{85} = 570.46 \pm 37.49 \text{ MHz}, \text{FWHM}_{87} = 580.97 \pm 47.40 \text{ MHz}.$$

These values are consistent with the theoretical Doppler widths calculated earlier (Eq. 6), though slightly larger. The excess broadening can be attributed to non-ideal gas behaviour in the vapour cell, including collisional broadening and imperfect alignment, which increase the effective velocity distribution beyond the ideal Maxwell–Boltzmann prediction.

Relative Amplitudes

The amplitudes of the Doppler-broadened features are approximately proportional to the relative number of atoms available in each hyperfine ground state. Rubidium contains two naturally occurring isotopes in the proportions $^{85}\mathrm{Rb}$ ~0.7217 and $^{87}\mathrm{Rb}$ ~0.2783.

Within each isotope, the ground-state populations scale with $N \propto (2F+1)$:

for ^{85}Rb the ratios of ground states population are: $2/5\ (F=2) : 3/5\ (F=3)$,
and for ^{87}Rb the ground states population ratios are: $3/8\ (F=1) : 5/8\ (F=2)$.

Combining the isotope abundances with these ground-state degeneracy ratios provides a first-order prediction for the relative absorption amplitudes. These predicted values agree closely with the observed Doppler-broadened peak heights.

A more complete description of relative line strengths is obtained from the Wigner–6j formalism, which determines transition probabilities ($F \rightarrow F'$). This treatment predicts slightly stronger transitions from the lower-F ground states. Incorporating this correction brings the expected and measured amplitudes into very good agreement.

Ground-State Magnetic-Dipole Constant A_s

For the ground-state $5^2S_{1/2}$ level, the total electronic angular momentum is $J = ½$, so the quadrupole coefficient is zero ($B = 0$). Equation 2 therefore reduces to

$$E_s(F) = \frac{A_s K_s(F)}{2} \qquad (15).$$

The two ground state energy levels F_{low} and F_{high} provide two simultaneous equations. When solved the Ground-State Magnetic-Dipole Constant is

$$A_s = 2\frac{\Delta E_s}{K_s(F_{high}) - K_s(F_{low})} \qquad (16).$$

(Here K refers to the quantity defined in Equation 3a).

Our measured values are:

$$A_s^{85} = 1005.7 \pm 15.1\text{MHz}, \quad A_s^{87} = 3416.1 \pm 5.1\text{MHz}.$$

These compare well with the published values:[4],[5]

$$A_s^{85} = 1011.9 \pm 15.1\text{MHz}, \quad A_s^{85} = 3417.3 \text{ MHz}.$$

Analysis of Doppler-Free Resonances

Lorentzian Linewidths of Doppler-Free Resonances

Each Doppler-free resonance was fitted with a Lorentzian profile (Eq. 8), whose full width at half maximum $\Gamma_{\mathrm{eff}}$is determined by the natural linewidth and power broadening (Eq. 9). Weighted averages based on individual fit uncertainties give the hyperfine Lamb-dip linewidths:

$$\Gamma_{\mathrm{eff}}^{87} = 11.25 \pm 0.01 \text{ MHz}, \quad \Gamma_{\mathrm{eff}}^{85} = 9.70 \pm 0.20 \text{ MHz}.$$

The crossover resonances are broader, with fitted linewidths:

$$\Gamma_{\times}^{85} = 13.79 \pm 0.03 \text{ MHz}, \;\; \Gamma_{\times}^{87} = 15.32 \pm 0.01 \text{ MHz}.$$

As expected from the statistical broadening mechanism described previously, the crossover linewidths are larger than the hyperfine Lamb dips by approximately a factor of $\sqrt{2}$, consistent with the contribution of two velocity classes $(\pm v_z)$to the crossover signal.
Our measured linewidth ratios agree with this prediction.

Determination of Excited-State Hyperfine Constants $A_p$and B_p

The Lorentzian fit of the Doppler-free resonances provides the hyperfine splitting energies in the excited $5^2P_{3/2}$state. According to the hyperfine energy equation

$$E(F') \; = \frac{A_p K(F')}{2} + \; b(F') \cdot B_p \qquad (1).$$

For each isotope and each ground-state level F, three excited-state energies $E(F')$ are measured for the allowed transitions with $\Delta F = -1, 0, +1$. From this a system of simultaneous equations can be solved to determine $A$and B

$$A_p = \frac{\Delta E_{12}(b_3 - b_1) - \Delta E_{13}(b_2 - b_1)}{\frac{1}{2}(K_2 - K_1)(b_3 - b_1) - \frac{1}{2}(K_3 - K_1)(b_2 - b_1)} \qquad (16),$$

$$B_p = \frac{\frac{1}{2}\Delta E_{13}(K_2 - K_1) - \frac{1}{2}\Delta E_{12}(K_3 - K_1)}{\frac{1}{2}(K_2 - K_1)(b_3 - b_1) - \frac{1}{2}(K_3 - K_1)(b_2 - b_1)} \qquad (17).$$

Applying the frequencies from the modelled Lorentzian hyperfine peaks gives:

$$A_p^{85} = 25.9 \pm 1.3 \text{ MHz}, \;\; A_p^{87} = 86.7 \pm 2.7 \text{ MHz},$$

$$B_p^{85} = 28.4 \pm 2.8 \text{ MHz}, \;\; B_p^{87} = 14.6 \pm 2.9 \text{ MHz}.$$

These results agree well with the accepted literature values (Steck 2025):[4],[5]

$$A_{\text{lit}}^{85} = 25.0354 \text{ MHz}, \;\; A_{\text{lit}}^{87} = 84.7185 \text{ MHz},$$

$$B_{\text{lit}}^{85} = 25.898 \;\; \text{MHz}, \;\; B_{\text{lit}}^{87} = 12.4965 \text{ MHz}.$$

The remaining differences fall within the experimental uncertainty dominated by sweep-rate nonlinearity and the finite signal-to-noise of the Doppler-free features, physical limitation in minimising the angle between pump and probe beams

Laser Sweep Non-Linearity Estimate

The time-to-frequency calibration assumes a linear laser sweep between adjacent Fabry–Pérot resonances, but diode lasers exhibit small nonlinear responses to current modulation.

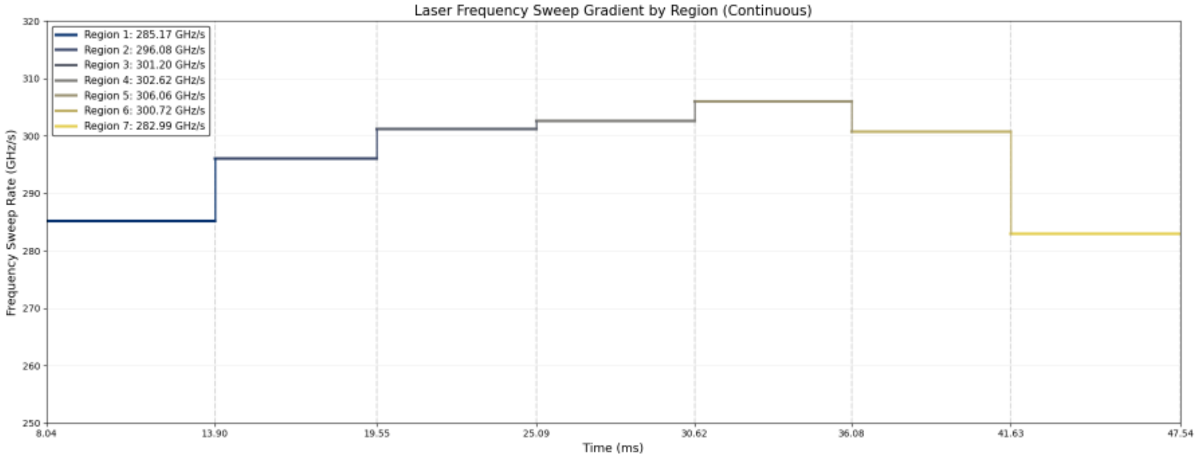

Figure 9 *illustrates laser sweep rate as an estimate using linear interpolation between cavity resonances*

This non-linearity was quantified using the hyperfine and crossover resonances: each crossover should lie exactly halfway between the two hyperfine transitions that form it, so any deviation from this midpoint provides a measure of the sweep-rate error. For each crossover, the fractional deviation ϵ was calculated as

$$\epsilon = \frac{f_\times - \frac{f_i+f_j}{2}}{\frac{f_i+f_j}{2}} \quad (18).$$

Using the weighted-deviation procedure described in Eqs. 14a–c, the combined result gives a sweep-rate non-linearity of approximately 1.5 ± 0.3%.

Conclusion

Doppler-free saturated absorption spectroscopy successfully resolved the hyperfine structure of the rubidium D_2 transition and enabled precise extraction of the magnetic-dipole and electric quadrupole constants for both naturally occurring isotopes. Calibration with a Fabry–Pérot cavity provided an accurate frequency scale within $\sim 1.5\%$ from frequency sweep non-linearity. Gaussian analysis of the Doppler-broadened transmission troughs was used to calculate the ground-state magnetic-dipole constants A_s, and Lorentzian analysis of the Doppler-free hyperfine peaks provided the excited-state constants A_p and B_p. All measured values agreed closely with established literature, confirming both the validity of the method and the reliability of the analysis procedure. Discrepancies were attributed to laser sweep nonlinearity, power broadening, and imperfect beam alignment, these effects could be limited in further experimentation

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